

Breeding the Mangrove Blue-eye

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The Mangrove Blue-Eye *Pseudomugil halophilus* is a relatively unknown member of the family Pseudomugilidae. It was probably found and kept by aquarists in the past, believing it to be a Pacific Blue-eye (*Pseudomugil signifer*), but it is now known to be a new species that has just been described. This species is found in the shallow, tidal, brackish waters of the east coast of Queensland and is clearly a schooling fish. They are very fast swimmers, using their agility to escape quickly at the slightest disturbance. This makes it very difficult to collect and it took a couple of hours to collect enough specimens.

The collected fish were juveniles and after the trip were placed in a 100 litre tank to grow in water conditions similar to those at the collection site, which were pH 7.7, salinity 32 ppt and 26°C. In the meantime, I searched for more information on these fish, but

there was virtually no information available, let alone any breeding reports. Further research revealed that this species has probably never been bred in captivity. So here's a challenge.

Based on aquarium observations, the Mangrove Blue-eye is a medium-large sized (5 cm total length), deep-bodied blue-eye. The body color is silverish with a few brown/yellow tinges and a row of brilliant blue or purple reflective scales along the posterior part of the lateral line. The first pointed dorsal fin and the rounded second dorsal fin are relatively short, yellowish and fringed with white. The ventral fins are small, transparent with a yellowish hue and a white fringe. The pectoral fins are transparent with some black accents and a partially white fringed. The yellow-colored anal fin is broad, long and partially white fringed. The white-margined caudal fin is transparent with little black stripes and a yellow conical spot.



Pseudomugil halophilus

Photo: Wim Heemskerk



Upper portion of Sand Hills Creek, Bowen Qld, habitat of the Mangrove Blue-eye.

Photo: Wim Heemskerk

Breeding tank setup

The Mangrove Blue-eye is a fast swimmer, so it can be expected that they will need some space to breed, which is why a 60x40x30 cm tank was chosen. The bottom was covered with a layer of sand mixed with some shell grit and an air driven sponge filter was used to filter the water and create a current. The water condition was determined based on the parameters at the collection site. Some mops were used to provide spawning material and a 6500K LED light was used as a light source. Water quality was maintained by changing approximately 10% of the water every other day. Some fish breeders prefer to breed with pairs or trios, others prefer to work with groups of animals to keep the genetic diversity as high as possible. In this case the breeding group consisted of 6 females and 7 males.

In order to get fish into good breeding condition, they need to be fed a varied diet. A wide range of foods is available, such as

frozen food, good quality flake/pellet food and of course live food. I prefer live food so I have fed a mixture of live brine shrimp (baby, juvenile and adult), mosquito larvae, flies and Dave Wilson's famous "pea and prawn" food recipe.

Breeding results

The Mangrove Blue-eye lives in brackish water and can cope well with varying salinity levels. They are unlikely to reproduce in pure sea- or pure freshwater, but somewhere in between. The trick is to find the sweet spot. The method used for clarification was to start at a salinity of 30 ppt and observe for 5 days for egg deposition. On day 6, the salinity was reduced by 5 ppt by changing the water. The sponge filter was cleaned and the cycle repeated. From the single trial here, it is clear that a salinity range between 10 ppt and 15 ppt gives the best results. The results are shown in Table 1.



Upper portion of Sand Hills Creek, Bowen Qld, habitat of the Mangrove Blue-eye.

Photo: Wim Heemskerk

Table 1. Results from breeding trials of Mangrove Blue-eye varying the aquarium water salinity.

Salinity	Egg count	Infertile/ moldy	Egg count	infertile/ moldy	Egg count	Infertile/ moldy	Egg count	Infertile/ moldy	Egg count	Infertile/ moldy
PPT	Day 1		Day 2		Day 3		Day 4		Day 5	
30.0	0		0		0		0		0	
25.0	0		0		0		0		0	
20.0	3	1	1	-	4	1	4	-	3	-
17.5	7	-	9	1	9	-	10	-	9	1
15.0	12	-	11	-	9	-	13	-	10	-
12.5	35	-	41	2	38	1	37	-	39	-
10.0	19	1	16	-	18	2	20	-	15	2
7.5	8	-	5	-	9	-	11	1	7	1
5.0	4	1	2	-	2	-	1	-	3	-

Each batch of eggs was stored in an aerated container with one litre of tank water at 26°C until hatching. No fungicide was added. The different salinities had no discernible effect on the development of the eggs, which are clear and approximately 2 mm in diameter. It takes the eggs quite a long time (18–21 days) to hatch. The newly hatched fry are relatively large and can be fed immediately with brine shrimp nauplii. The rapidly growing fry can be up to 1.5 cm long after 3 months.

The Mangrove Blue-eye adapts well to different salinity levels. Over time the salinity of the water has been lowered in order to test the fish's tolerance. Eventually they were kept in pure freshwater for several weeks, the fish had no problems with this but lost a little color. A small addition of salt brought the color back. The fish are now kept at a salinity of 3–4 ppt without any problems.



Juvenile Mangrove Blue-eye starting to colour up.

Photo: Wim Heemskerk



Pseudomugil halophilus

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